Mastering Strings in Python

Section 1: Learn

What are Strings in Python?

A **string** is a sequence of characters enclosed in **single (**'**), double (**"**), or triple (**''' **or** """**) quotes**.

Why are Strings Important?

- Used in **web development, data science, and automation**.
- **Text processing** is a core part of programming.
- Strings help in **storing and manipulating text**.

How Do Strings Work?

- Strings are **immutable** (cannot be changed once created).
- They can be **indexed and sliced**.
- Python provides many **built-in methods** to modify strings.

Interesting Anecdote:

Did you know that **every SMS you send is a string?**

Messaging apps process and format text using string operations, just like in Python!

Section 2: Practice

Creating and Accessing Strings

```
message = "Hello, Python!"
```

```
print(message[0])  # Output: H
print(message[-1])  # Output: !
```

Slicing a String

```
print(message[0:5])  # Output: Hello
print(message[:5])  # Output: Hello (starts from index 0 by default)
print(message[7:])  # Output: Python! (goes till end)
```

Modifying Strings with Built-in Methods

```
text = " python programming "


print(text.upper())  # Output: " PYTHON PROGRAMMING "
print(text.lower())  # Output: " python programming "
print(text.strip())  # Output: "python programming" (removes spaces)
print(text.replace("python", "Java"))  # Output: " Java programming "
```

Concatenation (Joining Strings)

```
first = "Hello"
second = "World"
result = first + " " + second


print(result)  # Output: Hello World
```

Looping Through a String

```
word = "Python"


for char in word:
    print(char)  # Prints each character in a new line
```

Real-World Example:

Imagine an **email validation system** that checks if an email contains @.

```
def validate_email(email):
    if "@" in email and "." in email:
        return "Valid Email"
    return "Invalid Email"


print(validate_email("test@gmail.com"))  # Output: Valid Email
print(validate_email("testgmail.com"))  # Output: Invalid Email
```

Try This Yourself:

1. Take a sentence and print **each word on a new line**.
2. Convert a **name input** to **uppercase**.

Section 3: Know More

Frequently Asked Questions

Q1: Can I change a character in a string?

No, strings are **immutable**. You must create a new string.

Q2: How do I check if a string contains a word?

Use the in keyword:

```
print("Python" in "I love Python")  # Output: True
```

Q3: What's the difference between strip(), lstrip(), and rstrip()?

- strip() removes spaces from **both sides**.
- lstrip() removes spaces from **left only**.
- rstrip() removes spaces from **right only**.

Q4: Can I store a multi-line string?

Yes! Use triple quotes (''' or """):

```
multiline = '''This is

a multi-line string'''
```

Q5: How do I find the length of a string?

Use len():

```
print(len("Python"))  # Output: 6
```

Strings are **powerful** and used everywhere. Keep practicing to master them!